Conditional probability

1

a $\frac{3}{4}$

b $\frac{5}{9}$

2

a $2, 4, 6, 8, 10, 12, 14, 16, 18, 20$

b $1, 2, 3, 4, 6, 8, 12, 16$

c 6

d $\frac{3}{5}$

3

a

i 260

ii 237

iii 111

iv 208

b $\frac{208}{319}$

4

a 28 combinations

b $\frac{1}{4}$

c $\frac{1}{4}$

5

a

	1	2	4	6
1	1	2	4	6
2	2	4	8	12
3	3	6	12	18
4	4	8	16	24
5	5	10	20	30
6	6	12	24	36

b $\frac{1}{2}$

c $\frac{3}{10}$

d $\frac{1}{2}$

6

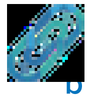
a $\frac{9}{19}$

b $\frac{4}{19}$

c $\frac{3}{19}$

7

a $\frac{1}{6}$



b $\frac{1}{3}$

8

a $\frac{1}{12}$

b $\frac{1}{3}$

c $\frac{1}{4}$

9

a $\frac{1}{9}$

b $\frac{5}{9}$

c $\frac{1}{3}$

10

a

	2	4	5
3	5	7	8
7	9	11	12

b $\frac{2}{5}$

c 1

11

a Patricia is chosen second, after Aaron has been chosen first.

b Greater than

12 0.4

13 $\frac{1}{3}$

14

a No

b Yes

c Yes

d No

e Yes

f No

15

a 3

b $\frac{47}{50}$

c $\frac{3}{10}$

16

a $\frac{16}{69}$

b 10

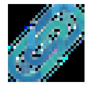
c $\frac{5}{13}$

d $\frac{2}{3}$

17 a

i 0.23 ii 0.73 iii $\frac{27}{50}$ iv $\frac{23}{56}$

v $\frac{17}{50}$



i 0.1 ii 0.6 iii $\frac{4}{5}$ iv $\frac{1}{5}$

v $\frac{1}{5}$

18

a

i $x = 426$ ii $y = 398$ iii $z = 154$

b $\frac{199}{489}$

c $\frac{77}{276}$

19

a 0.7 b Yes c 0.5

20

a

i $x = 40$ ii $y = 400$ iii $z = 200$

b 360

c $\frac{2}{3}$

21

a 0.18 b 0.45 c 0.38 d 0.64

e 0.9

22

a $\frac{4}{13}$ b $\frac{2}{3}$ c $\frac{1}{2}$ d $\frac{4}{9}$

23 0.1875

Independent events

24 0.2

25

a Yes

b No

26

a 0.42

b 0.12

c 0.18

d 0.18

e Independent

27

a 0.3

b 0.12

c 0.42

d 0.18

e Independent

28

a 0.8

b 0.05

c $P\left(B\right) = x + 0.05$

d $x = 0.2$

e 0.25

29

a $x = 0$

b $x = \frac{3}{10} \frac{1}{5}$

c $x = 0.3$

Applications

30

a 177

b $\frac{177}{289}$

c $\frac{65}{242}$

31

a 346

b 12.5%

c $\frac{74}{173}$

d $\frac{7}{8}$

e $\frac{29}{173}$

32

a 142

b 98

c 65.79%

d 51.02%

e 15.65%

33

a 33.33%

b $\frac{10}{27}$

c $\frac{50}{77}$

d $\frac{47}{77}$

34

a $\frac{131}{976}$

b $\frac{1}{16}$

c $\frac{250}{1881}$

d $\frac{1681}{1690}$

35

a $\frac{47}{74}$

b $\frac{53}{74}$

c $\frac{1}{2}$

d $\frac{29}{42}$

36

a $\frac{98}{1153}$

b $\frac{5274}{5765}$

c $\frac{496}{497}$

d $\frac{248}{493}$

37

a

	Democrat (next election)	Republican (next election)	Independent (next election)	Total
Democrat (last election)	26	41	3	70
Republican (last election)	15	74	1	90
Independent (last election)	9	25	6	40
Total	50	140	10	200

b $\frac{1}{6}$

c $\frac{2}{5}$

d $\frac{23}{32}$

38

a

	Believe in Santa	Do not believe in Santa	Total
Believe in Tooth Fairy	31	10	41
Do not believe in Tooth Fairy	49	10	59
Total	80	20	100

b 38.75%

c $\frac{31}{90}$

d $\frac{10}{41}$

39

a

	E-Reader	Not E-Reader	Total
Paperback	60	84	144
Not Paperback	20	36	56
Total	80	120	200

b $\frac{21}{50}$

c 42%

40

a

	Watched the movie	Didn't watch the movie	Total
Read the book	39	58	97
Didn't read the book	65	4	69
Total	104	62	166

b

i $\frac{65}{166}$

ii $\frac{39}{97}$

iii $\frac{29}{31}$

41

a $\frac{2}{3}$

b $\frac{2}{5}$

42

a 0.66

b $\frac{4}{11}$

43

a 25

b $\frac{17}{87}$

c $\frac{25}{42}$

d $\frac{17}{27}$